HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** microservices for enterprise systems, implementing **REST APIs** and handling cross-service communication through **JPA**.
- Engineered **Python scripts** to automate data extraction, transformation, and loading processes, improving reporting efficiency for a financial services client.
- Managed production environments for enterprise applications using **Splunk** and **Dynatrace**, responding to P1/P2 incidents and ensuring SLA adherence.
- Designed and built containerised applications using **Docker** for telemetry collection and local data processing on **Linux**-based edge systems, ensuring secure communication with cloud platforms.
- Enhanced **CI/CD pipelines** using **Jenkins** and **Git** for automated testing and deployment of software components, reducing manual intervention by **30%**.
- Collaborated with data science and platform teams to integrate and optimise **machine-learning models** on edge devices, focusing on telemetry data processing and lifecycle management.
- Applied **secure-by-design principles**, including certificate-based authentication and secrets management, to protect device identity and ensure encrypted communication channels.

SKILLS

Languages – Java, Python, C#, SQL, R

Frameworks & Libraries – Spring Boot, JPA, Hibernate, pandas, NumPy, scikit-learn, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Linux

Databases – PostgreSQL, MySQL, Oracle, MongoDB

IoT & Protocols – MQTT, gRPC, OPC UA, HTTP/REST, WebSockets, AMQP

Tools & Platforms – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau

PROJECTS

Edge Device Telemetry and Processing Platform

- Developed containerised **Python** services deployed via **Docker** on **Linux** for real-time telemetry ingestion from IoT devices, handling approximately **1,500 events** per second.
- Implemented local data buffering and pre-processing logic using **pandas** to reduce network bandwidth and prepare data for secure transmission to cloud endpoints via **MQTT**.
- Integrated secure communication protocols, including TLS encryption and device certificate validation, to ensure data integrity and prevent unauthorized access.

ML Model Deployment and Monitoring Pipeline

- Engineered **Python** scripts to automate the packaging and deployment of **machine-learning models** to edge devices, integrating with a **CI/CD pipeline** managed by **Jenkins**.
- Built data pipelines to collect performance metrics and inference logs from deployed models, feeding into an operational dashboard for anomaly detection and lifecycle management.
- Collaborated with data scientists to refine the MLOps tooling, enhancing the reliability and scalability of model updates across the device fleet.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)